2 ENKRIPSI CITRA LOGO DENGAN INVERS MATRIKS 3x3

a. Matriks kunci A

subs a=0, b=2 ke dalam template matriks

$$A = \begin{bmatrix} a+1 & a & 2 \\ 1 & a+2 & b \\ b & 1 & a+3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 2 \\ 1 & 2 & 2 \\ 2 & 1 & 3 \end{bmatrix}$$

hitung det(A) [ekspansi kofaktor b1)

$\det(A) = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}$

$C_{11} = +\det\begin{bmatrix} 2 & 2 \\ 1 & 3 \end{bmatrix} = (2\times3)-(2\times1) = 6-2 = 4$

$C_{12} = -\det\begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} = -((1\times3)-(2\times2)) = -(3-4) = 1$

$C_{13} = +\det\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} = (1\times1)-(2\times2) = 1-4 = -3$

$\det(A) = (1\times4)+(0\times1)+(2(-3)) = 4+0-6 = -2$

$\det(A) = -2 \neq 0 \rightarrow$ matriks invertible ✓
enkripsi/deskripsi valid

b. Tabel 9 Matriks Minor M_{ij}

i	j	submatriks	Nilai minor
1	1	det [[2,2],[1,3]]	4
1	2	det [[1,2],[2,3]]	-1
1	3	det [[1,2],[2,1]]	-3
2	1	det [[0,2],[1,3]]	-2
2	2	det [[1,2],[2,3]]	-1
2	3	det [[1,0],[2,1]]	1
3	1	det [[0,2],[2,2]]	-4
3	2	det [[1,2],[1,2]]	0
3	3	det [[1,0],[1,2]]	2

c. Matriks kofaktor C dan Adjoin Adj(A)

Tanda kofaktor : $C_{ij} = (-1)^{(i+j)} \times M_{ij}$

$$C = \begin{bmatrix} +4 & +1 & -3 \\ +2 & -1 & -1 \\ -4 & 0 & +2 \end{bmatrix}$$

Adj(A) = C^T (transpose matriks kofaktor) :

$$Adj(A) = \begin{bmatrix} 4 & 2 & -4 \\ 1 & -1 & 0 \\ -3 & -1 & 2 \end{bmatrix}$$

$A^{-1} = \left(\frac{1}{\det(A)}\right) \times adj(A) = \left(\frac{1}{-2}\right) \times adj(A) :$

$$A^{-1} = \begin{bmatrix} -2 & -1 & 2 \\ -1/2 & 1/2 & 0 \\ 3/2 & 1/2 & -1 \end{bmatrix}$$

d. kode matlab.

e. Studi kasus Enkripsi - Deskripsi Blok Pixel.

subs a=0, b=2 ke blok piksel P :

$$P = \begin{bmatrix} 120 & 50 & 200 \\ 80 & 2 & 12 \\ 32 & 90 & 190 \end{bmatrix}$$

Enkripsi E = A×P (perhitungan baris demi baris) :

Baris 1A = [1, 0, 2]

$E_{11} = (1\times120)+(0\times80)+(2\times32)$
$= 120+0+64 = 184$

$E_{12} = (1\times50)+(0\times2)+(2\times90)$
$= 50+0+180 = 230$

$E_{13} = (1\times200)+(0\times12)+(2\times190)$
$= 200+0+340 = 540$

Baris 2A = [1, 2, 2]

$E_{21} = (1\times120)+(2\times80)+(2\times32)$
$= 120+160+64 = 344$

$E_{22} = (1\times50)+(2\times2)+(2\times90)$
$= 50+4+180 = 234$

$E_{23} = (1\times200)+(2\times12)+(2\times190)$
$= 200+24+340 = 564$

Baris 3A = [2, 1, 3]

$E_{31} = (2\times120)+(1\times80)+(3\times32)$
$= 240+80+96 = 416$

$E_{32} = (2\times50)+(1\times2)+(3\times90)$
$= 100+2+270 = 372$

$E_{33} = (2\times200)+(1\times12)+(3\times190)$
$= 400+12+570 = 922$

$$E = \begin{bmatrix} 184 & 230 & 540 \\ 344 & 234 & 564 \\ 416 & 372 & 922 \end{bmatrix}$$

$P' = A^{-1} \times E = A^{-1} \times (A \times P) = (A^{-1}A) \times P = I \times P = P$ ✓

karena $A^{-1}A = I$ (matriks identitas), maka $P' = P$ untuk setiap blok piksel.

f. kode matlab.

Dipindai dengan CamScanner